

10 June 2026

Taiwan Technology

Focus stocks

ASE	3711 TT
Rec	O-PF
Market cap	US\$72.6bn
3M ADV	US\$363.5m
Price	NT\$539
Target	NT\$161→670
Up/downside	+24%

KYEC	2449 TT
Rec	O-PF
Market cap	US\$10.5bn
3M ADV	US\$331.9m
Price	NT\$272
Target	NT\$142→360
Up/downside	+32%

Powertech	6239 TT
Rec	HLD→O-PF
Market cap	US\$7.5bn
3M ADV	US\$155.8m
Price	NT\$303
Target	NT\$125→370
Up/downside	+22%

SinoPac analysts:
Astoria Chen, SinoPac

www.clsa.com

OSAT rewired

AI complexity driving higher requirements for packaging and testing

CoWoS demand is soaring as the performance requirements of AI accelerators is driving a demand shift to advanced packaging to realise system-level integration. Advanced packaging needs and tight CoWoS capacity are driving OSAT participation and the adoption of alternative technologies like CoPoS/FOPLP and SoIC. The transition to chiplet-based AI architectures and growing AI chip shipments also increases testing complexity, thereby boosting its importance. We view OSATs as the key beneficiaries of this trend, and assume coverage of ASE, KYEC, and Powertech at Outperform.

AI changing chip requirements

- Unlike traditional chips which mainly focused on transistor scaling and cost efficiency, AI accelerators are simultaneously demanding larger die size, higher memory bandwidth, and faster interconnect speed, driving a demand shift to advanced packaging to realise system-level integration.
- AI semiconductors are more complex and expensive, using the most advanced node process, so the tolerance for failure across the supply chain is declining significantly.

Packaging becoming performance-critical

- At advanced nodes, scaling large dies is limited by manufacturing costs and yield, making CoWoS package-level integration of logic and HBM crucial for supporting AI workloads with higher interconnect density.
- TSMC's CoWoS capacity has remained extremely tight despite its accelerated expansion, which creates a more direct opportunity for OSAT (outsourced semiconductor assembly and test) players such as Amkor and ASE to participate in the AI advanced packaging supply chain.
- As growing package sizes make large silicon interposers less efficient due to wafer utilisation, yield, warpage, and cost challenges, CoPoS/FOPLP and SoIC are emerging to improve scalability, cost efficiency, and compute density for AI accelerators.

Testing is another hidden bottleneck

- The move to chiplet-based AI architectures increases potential failure points and necessitates greater pre- and post-assembly testing, setting the stage for longer test time and more complex test flows.
- With higher AI chip shipments, longer test time, more test insertions, and increasing adoption of burn-in and system-level test, we believe testing will become an increasingly important growth driver for the leading testing providers such as KYEC.

Assuming coverage of OSAT names

- **ASE (O-PF):** We expect ASE to sustain its strong LEAP platform momentum in 2026 and beyond, benefiting from oS outsourcing, TSMC spillover orders, and growing AI-driven advanced packaging and testing demand to deliver long-term growth. We assume coverage at O-PF with a TP of NT\$670 (previously NT\$161).
- **KYEC (O-PF):** We view KYEC as one of the best AI chip proxies in Taiwan OSAT, with high GPU and ASIC exposure and dominant positions in major AI chip projects, and we expect structural AI testing growth to drive profitability as Rubin concerns fade with ramp-up from 3Q26. We assume coverage at O-PF with a TP of NT\$360 (previously NT\$142).
- **Powertech (O-PF):** We see Powertech entering a revenue and margin upcycle fueled by a memory supercycle, AI testing demand, and emerging FOPLP and optical packaging opportunities. We assume coverage at O-PF (previously Hold) with a TP of NT\$370 (previously NT\$125).

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Growing importance of packaging and testing

Moore's Law scaling faces growing limits

Transistor count on chip keeps increasing

AI changing chip requirements

AI is driving a structural shift in semiconductor scaling logic. While transistor scaling remains important, increasingly large and complex AI accelerators are pushing the limits of traditional monolithic design, making system-level integration, memory bandwidth and interconnect efficiency increasingly critical determinants of compute performance. At the same time, rapidly rising chip value and deployment complexity are significantly increasing reliability requirements, making packaging and testing increasingly important bottlenecks within the AI semiconductor supply chain.

AI chip design drives a shift beyond traditional scaling

Over the past several decades, Moore's Law has remained the key driver of semiconductor performance improvement through continuous transistor scaling. It enables the creation of smaller, faster, and more productive devices, transforming various industries such as computing, telecommunications, and consumer electronics. While advanced nodes continue to progress into the 3nm and upcoming 2nm era with the emergence of AI accelerators, the pace of scaling is becoming increasingly constrained by both physical and economic limitations. Challenges including reticle limits, power leakage, and thermal dissipation, and rapidly escalating fabrication costs have significantly reduced the efficiency of traditional monolithic scaling.

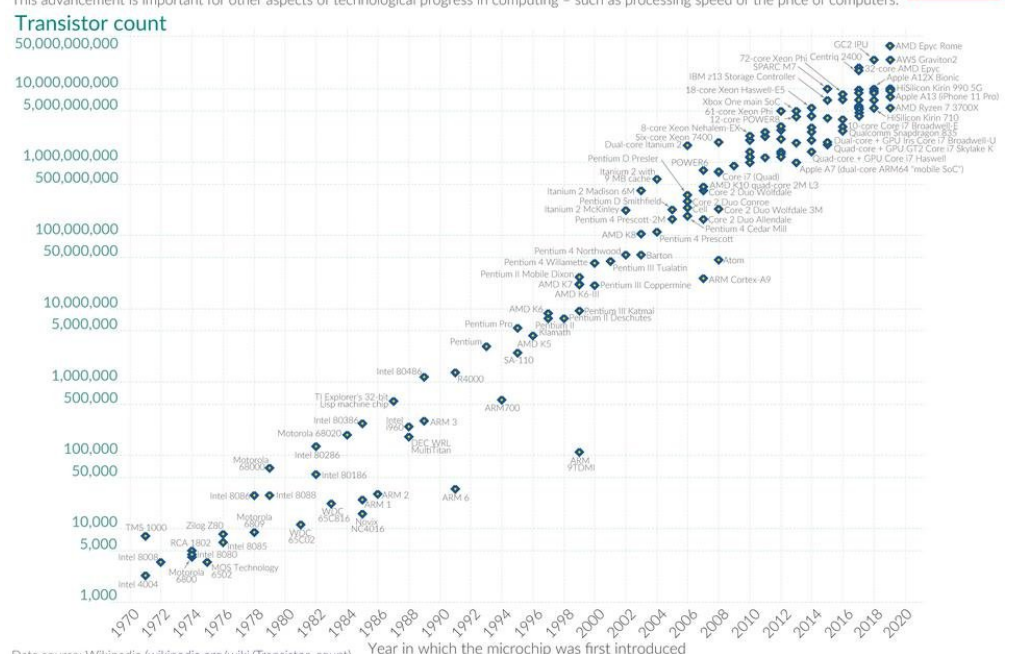
Figure 1

Transistor count on chip

Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Our World in Data



Source: SinoPac, Our World in Data

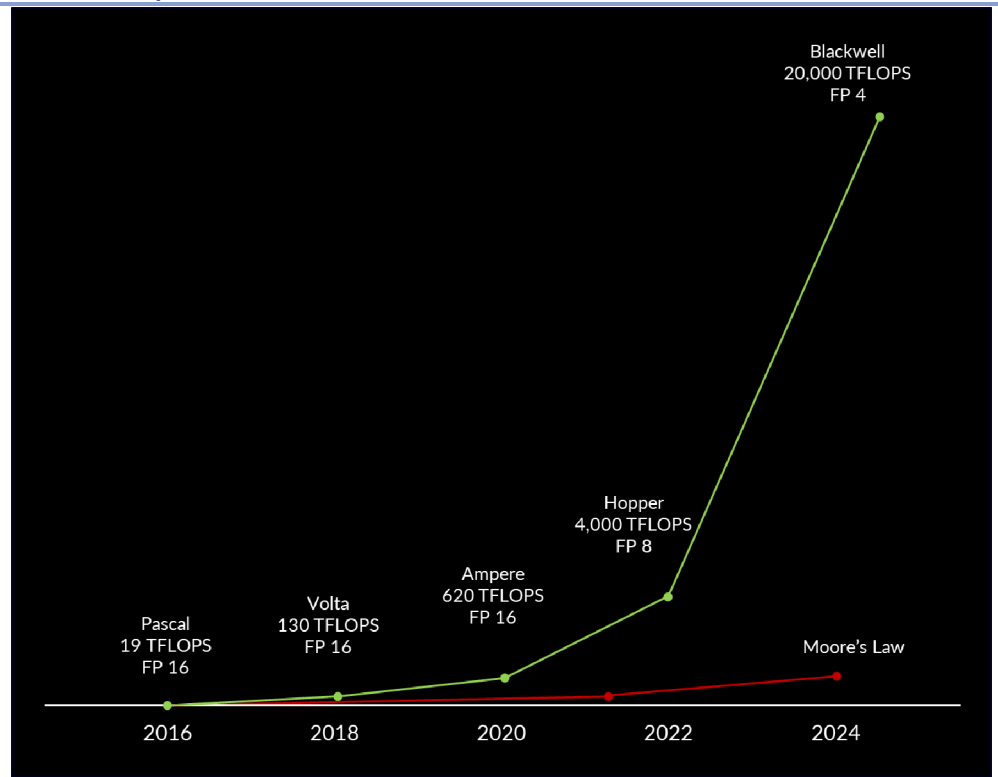
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AI chips becoming increasingly powerful

Figure 2

Nvidia AI GPU products



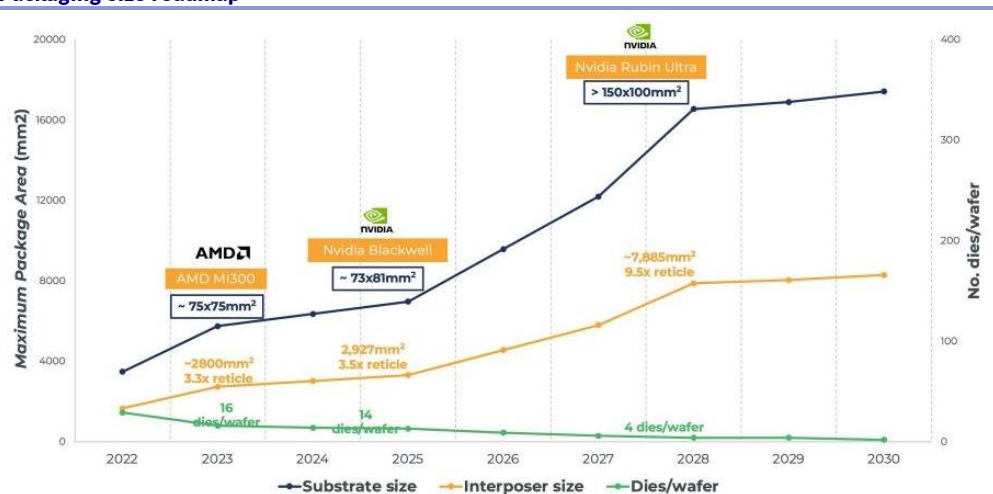
Source: SinoPac, Nvidia

AI chip architecture getting more complex

As AI semiconductor architectures become increasingly heterogeneous and modularised, system-level integration is becoming more important across the semiconductor value chain. Unlike traditional chips that mainly focused on transistor scaling and cost efficiency, AI accelerators are simultaneously demanding larger die size, higher memory bandwidth, faster interconnect speed and better power efficiency, significantly increasing overall system complexity.

Figure 3

Packaging size roadmap



Source: SinoPac, Yole

Advanced packaging evolving towards larger packaging size

Interconnect limits challenge advanced chip scaling

With process nodes continuing to migrate towards the 3nm and upcoming 2nm era, transistor density continues to improve, but interconnect pitch scaling is advancing at a slower pace. As a result, maintaining higher transistor counts and I/O density increasingly requires larger silicon area, pushing AI chips closer to the reticle limit and making further monolithic scaling less economical. This trend can be seen in Nvidia's AI GPU roadmap, where system complexity and die count continue to increase from Hopper's monolithic design to Blackwell's dual-die design and further into Rubin and Feynman architecture.

Figure 4

Nvidia GPU roadmap

	H200	B200	B300	Rubin	Rubin Ultra	Feynman
Launch	2Q24	4Q24	3Q25	2H26	2H27	2H28
Architecture	Monolithic	Dual-die	Dual-die	Dual-die + 2* I/O	(Dual die + 2* I/O)*2	2+2 stacking
Process node	N4P	N4P	N4P	N3P	N3P	A16
HBM (GB)	141GB (HBM3e)	192GB (HBM3e)	288GB (HBM3e*8)	288GB (HBM4*8)	1024GB (HBM4e*16)	HBM4e?
Memory Bandwidth (TB/s)	4.8	8	8	22	32	TBD
Packaging	CoWoS-S	CoWoS-L	CoWoS-L	CoWoS-L	CoWoS-L	CoWoS-L + SoIC
Interconnect	NVLink 900GB/s	NVLink 1800GB/s	NVLink 1800GB/s	NVLink 3600GB/s	NVLink 3600GB/s	NVLink 7200GB/s
TDP(W)	700	1000	1000	1800	2500	5920
Networking	400Gb CX7	400Gb CX7	800Gb CX8	1600Gb CX9	1600Gb CX9	3200Gb CX10

Source: SinoPac, Nvidia

AI workloads driving compute performance requirements

At the same time, AI workloads are becoming increasingly bandwidth- and latency-sensitive, making memory access and die-to-die communication critical determinants of overall compute performance. Hyperscalers are therefore increasingly adopting chiplet-based architectures, where compute dies utilise the most advanced process nodes while I/O dies remain on last generation nodes to optimise manufacturing cost and yield.

Figure 5

Google TPU roadmap

Code name	Carmano	Mustang	Axe (Trillium)	Hammer (Ironwood)	Zebrafish (A5921)	Sunfish	Humufish (A5922)	Whalefish (Sunfish Ultra)	Icefish
Market name	TPU v5e	TPU v5p	TPU v6e	TPU 7x	TPU 8t	TPU 8i	TPU 9t?	TPU v9i?	TPU v10
Process	CoWoS-S	CoWoS-S	CoWoS-S	CoWoS-S	CoWoS-S	CoWoS-L	EMIB-T	CoWoS-L	EMIB-T?/CoWoS-L?
Design Service	Broadcom	Broadcom	Broadcom	Broadcom	MediaTek	Broadcom	Google + MediaTek	Broadcom	TBD, MediaTek I/O?
Compute die node	N5 x1	N5 x1	N5 x2	N3E x2	N3E x1	N3P x2	N2 x4	N3 x4	
Other logic die node							N3 x4		
I/O die node	N7 x1	N7 x1		N5 x1	N4 x1	N4 x1	N3 x4	N4 x2	
HBM	HBM2E x2	HBM2E x6	HBM3E x2	HBM3E x8	HBM3e x6	HBM3e x8	HBM4e *12	HBM3e *16	
MP schedule	2Q22	2Q23	2Q24	2H25	4Q26	2H26	4Q27/1Q28	2027?	2029/30?

Source: SinoPac, Google

AI chip defects have higher economic impact

AI chips are too expensive to fail

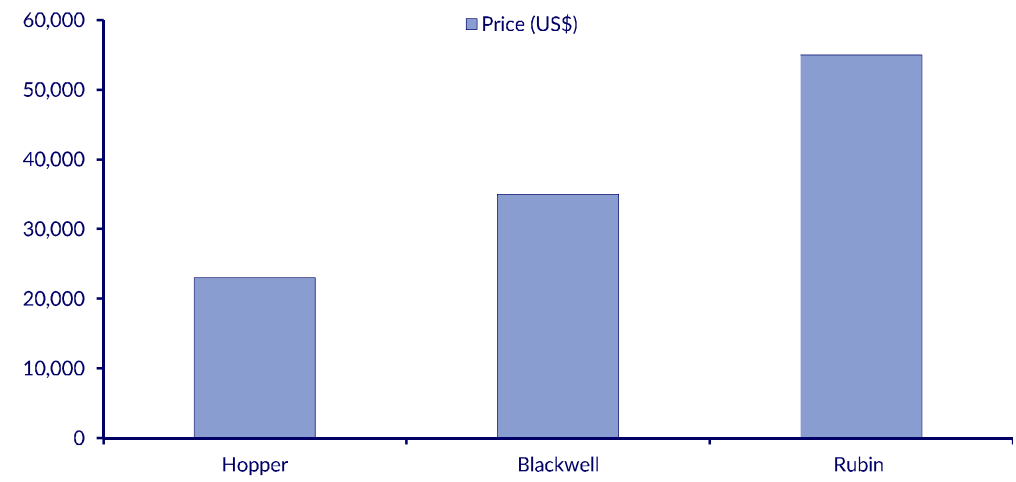
AI semiconductors are more complex and expensive as they use the most advanced node process, so the tolerance for failure across the supply chain is declining significantly. Compared with traditional semiconductors, AI GPUs and ASICs carry

GPU prices increasing

substantially higher silicon content, larger die size, more advanced packaging layers and significantly higher HBM content, resulting in a much higher package value per unit. That means a single defect may result in the loss of multiple compute and memory dies simultaneously, significantly increasing the economic impact of failure.

Figure 6

Nvidia GPU price comparison



Source: SinoPac, Nvidia

Rising testing demand driven by higher performance requirements

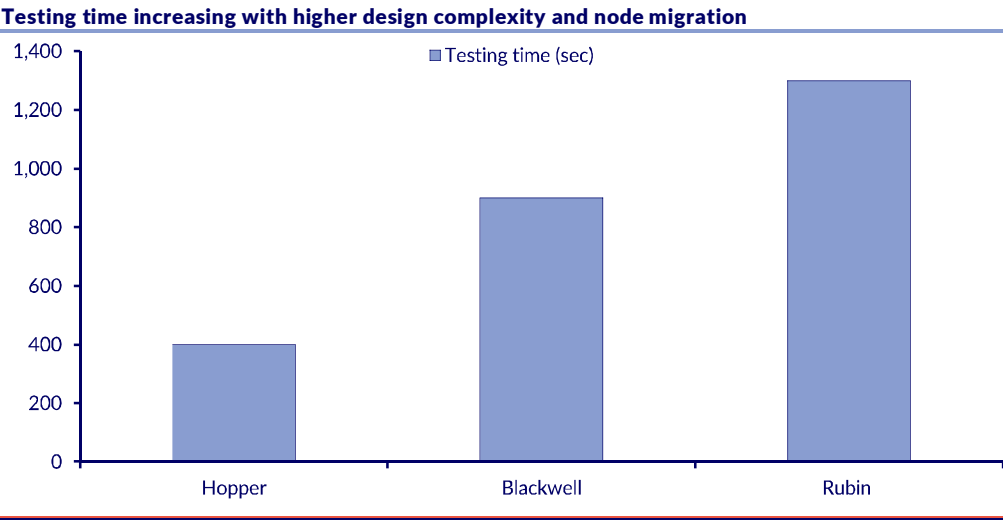
At the same time, AI workloads are pushing chips to operate at substantially higher power consumption and system complexity, increasing the importance of thermal stability, memory integrity and die-to-die interconnect reliability. As a result, validation requirements are becoming increasingly strict, with longer testing time and deeper testing coverage increasingly required for AI semiconductors. Testing is therefore evolving beyond traditional quality assurance towards system-level reliability validation.

AI chip testing shifting to supply critical point

In addition, AI semiconductors are increasingly deployed at rack- and cluster-level scale, where the cost of downtime or field failure can materially exceed the cost of testing itself. Hyperscalers are therefore placing greater emphasis on burn-in, system-level testing (SLT) and overall reliability management to minimise deployment risk. As AI semiconductors become increasingly valuable and mission-critical, testing is evolving from a downstream manufacturing step into a strategic bottleneck within the semiconductor supply chain.

Longer testing aligns with AI chip roadmap

Figure 7



Source: SinoPac

Figure 8

Comparison of testing process			
Test Stage	Level	Applies to Chip Types	Purpose
Chip Probing	Wafer level	All chip types	Basic electrical check of die at wafer level; items involve lower power
Final Testing	Device level	All chip types	Full check of packaged chip to meet specs; test time scales with complexity, process node, and feature count
Burn-In Testing	Device level	Mainly HPC, server, mission-critical chips	Stress test at high temp/voltage to find early failures; adds hours to test time
System-Level Testing (SLT)	Device level	Mainly HPC, server, mission-critical chips	Test chip in a near-real or actual system environment; costly/time-consuming

Source: SinoPac

Packaging becoming performance-critical

Advanced packaging has become the most immediate bottleneck and value-creation point in the AI accelerator supply chain. CoWoS remains the mainstream solution today, as it enables high-density integration of logic dies and HBM to meet AI's bandwidth and latency requirements. However, rising accelerator complexity is pushing packaging requirements well beyond conventional 2.5D scaling, with larger interposers, more HBM stacks, higher die counts and increasing system-level integration all driving capacity tightness and technical complexity. While TSMC should remain the dominant platform provider, its capacity constraints and process outsourcing are creating more direct opportunities for OSAT players such as ASE. Longer term, as CoWoS scaling becomes increasingly challenging, SiIC-based 3D stacking and panel-level solutions such as CoPoS/FOPLP are likely to become important extensions of the AI packaging roadmap, expanding both the technical scope and addressable market for advanced packaging for both TSMC and OSAT vendors.

CoWoS capacity tightening
driven by strong AI demand

CoWoS enables the first wave of AI scaling

Since the AI cycle took off in 2023, Nvidia's H100 has brought advanced packaging into the spotlight, with CoWoS becoming the most well-known solution and the mainstream packaging architecture for current-generation AI accelerators. By enabling high-density integration between logic dies and HBM, CoWoS directly addresses the bandwidth and latency requirements of AI workloads. However, with AI accelerator demand ramping rapidly, CoWoS capacity also became one of the key supply bottlenecks, alongside leading-edge wafers and HBM. For a prolonged period, packaging was no longer a back-end consideration, but a direct constraint on AI accelerator output.

Performance requirements
driving change in packaging
design

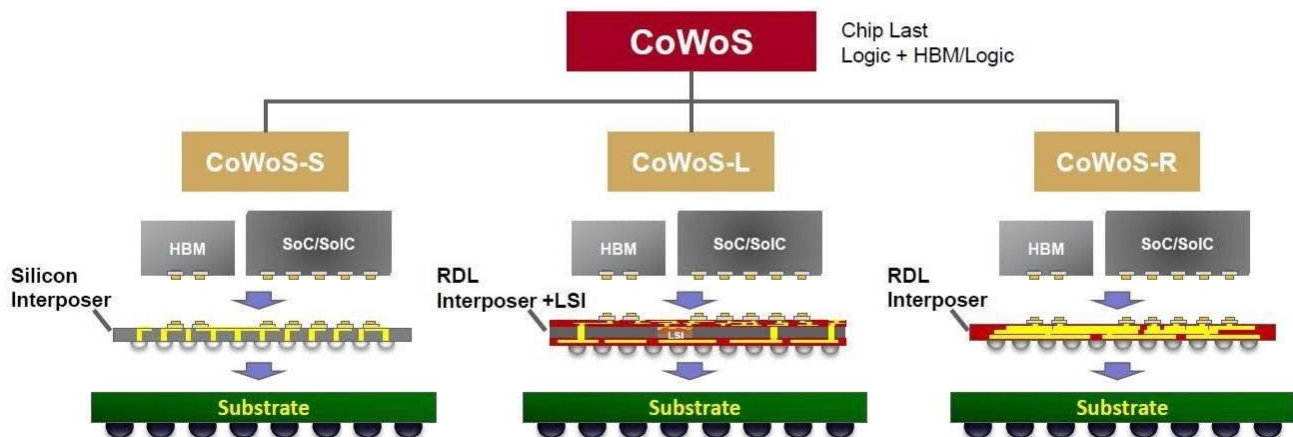
The rise of CoWoS reflects a broader shift in AI chip design, where performance improvement is no longer driven by transistor scaling alone. At 5nm and 3nm, larger monolithic dies face increasing constraints from reticle limits, yield loss and rising wafer costs, while AI workloads require massive memory bandwidth supported by HBM. Through its silicon interposer, CoWoS enables logic dies and HBM stacks to be integrated within the same package with much higher interconnect density than traditional substrate-based packaging, making package-level integration a key enabler of further accelerator scaling. This marks the first clear stage where advanced packaging has evolved from a back-end manufacturing step into a performance-critical and capacity-critical part of the AI semiconductor supply chain.

Figure 9

TSMC CoWoS technology

CoWoS® Platform for HPC AI Applications

- A versatile 2.5D packaging technology for heterogeneous chiplet integration



Source: SinoPac, TSMC

CoWoS demand growth surpassing capacity expansion

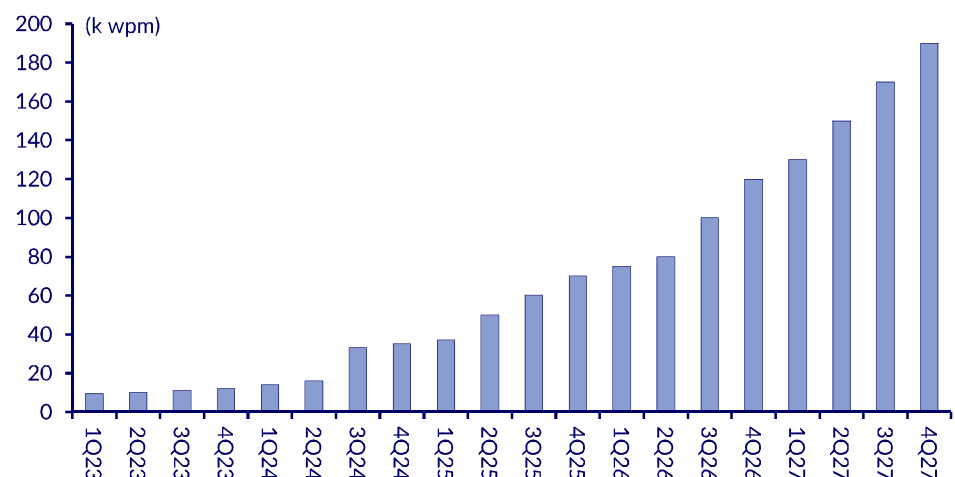
~190kwpm by end-2027SP

Outsourcing creates opportunities beyond TSMC

As almost all AI accelerators, including GPUs, ASICs, and even server CPUs require CoWoS or CoWoS-like advanced packaging, TSMC's CoWoS capacity has remained extremely tight despite its accelerated expansion. We expect TSMC's CoWoS capacity to reach around 190kwpm by end-2027, but believe this is still unlikely to fully satisfy rapidly growing AI accelerator demand. At the same time, for better cost efficiency and capacity flexibility, TSMC is currently focusing its internal expansion mainly on the CoW process, while outsourcing part of the oS process to partner OSATs such as ASE. This creates a more direct opportunity for OSAT players to participate in the AI advanced packaging supply chain, even as TSMC remains the dominant CoWoS provider.

Figure 10

TSMC CoWoS capacity



Source: SinoPac

OSATs benefiting from spillover orders from TSMC

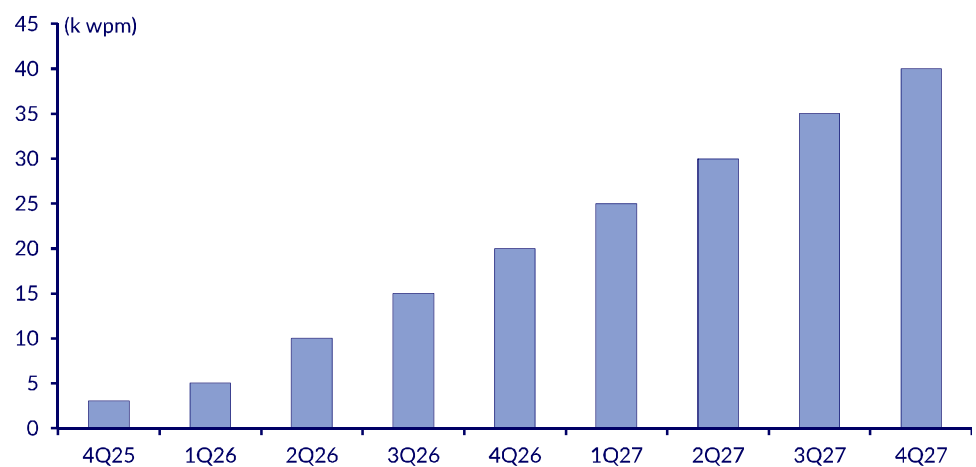
Full process capacity to reach 30-40k by end-2027SP

Packaging size increasing

OSATs are also actively expanding CoWoS-like advanced packaging capacity to capture spillover demand, such as Amkor and ASE. We expect ASE's full-process capacity to reach 15-20k/month by end-2026 and 30-40k/month by end-2027, supported by demand from large clients such as Nvidia, AMD, Marvell and Broadcom. Given tight supply at TSMC, design houses are gradually shifting non-accelerator CoWoS demand to OSAT vendors, including CPUs, networking ASICs and certain consumer-related projects. As these OSAT solutions are already silicon-proven and offer a more readily scalable alternative, ASE should be a key beneficiary in Taiwan, with solid client demand into 2026-27.

Figure 11

ASE CoWoS-like capacity forecast (full process)



Source: SinoPac

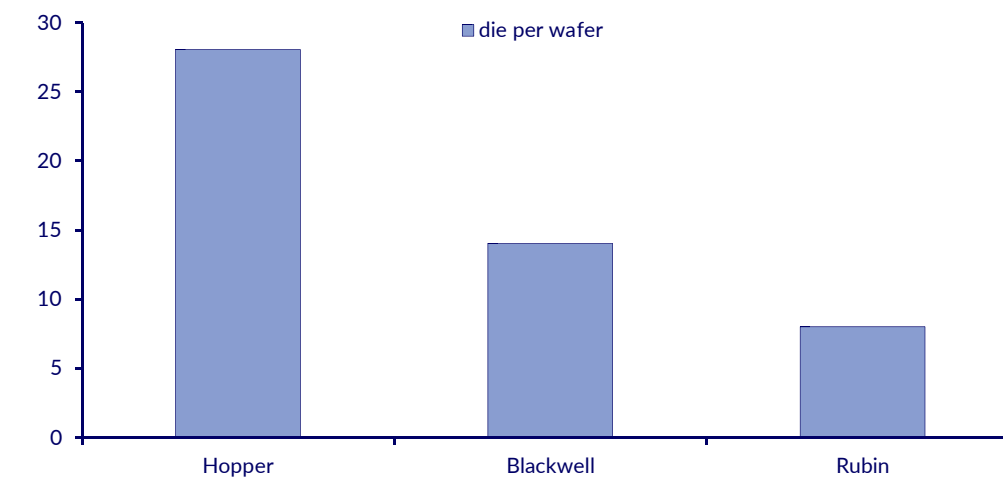
Further scaling requirements driving emergence of new solutions

As AI accelerator specifications continue to rise, design requirements are moving beyond the integration of one large logic die and several HBM stacks. Logic die size is increasing, HBM count is rising, and design houses are looking to integrate more compute dies, I/O dies and memory within the same package. TSMC is still scaling CoWoS to support this trend, with interposer size moving from 3.3-reticle with 8x HBM3 to 5.5-reticle with 12x HBM3E/4, and further towards more than 14-reticle with 24x HBM by 2029.

GPU die per wafer declining

Figure 12

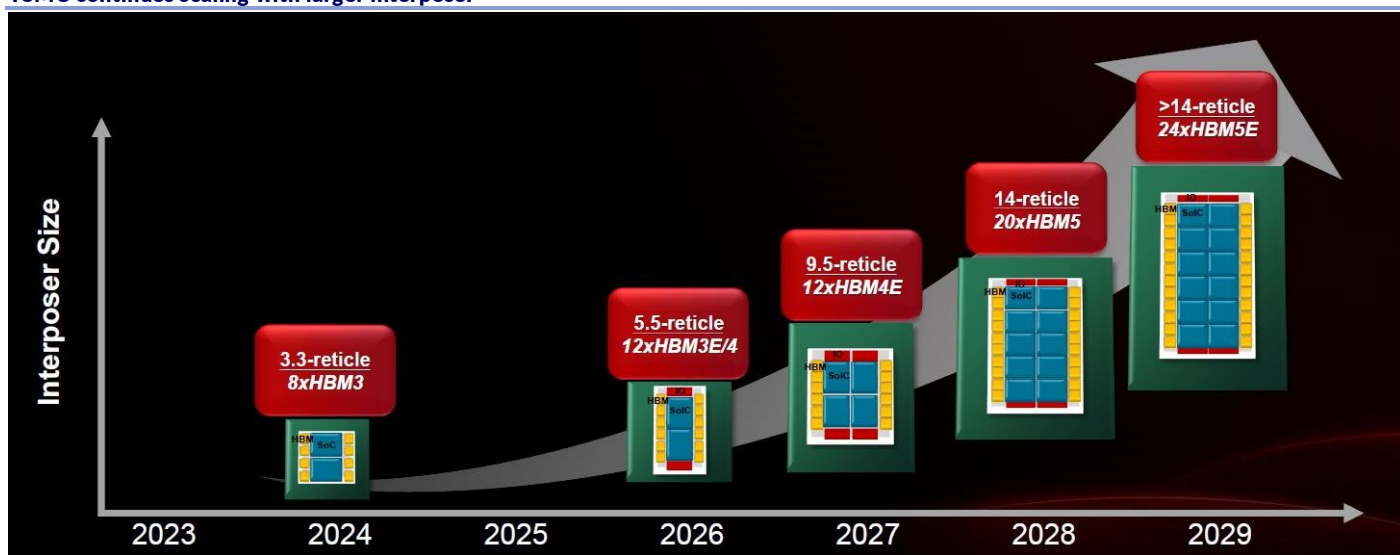
Nvidia GPU die per CoWoS wafer



Source: SinoPac

Figure 13

TSMC continues scaling with larger interposer



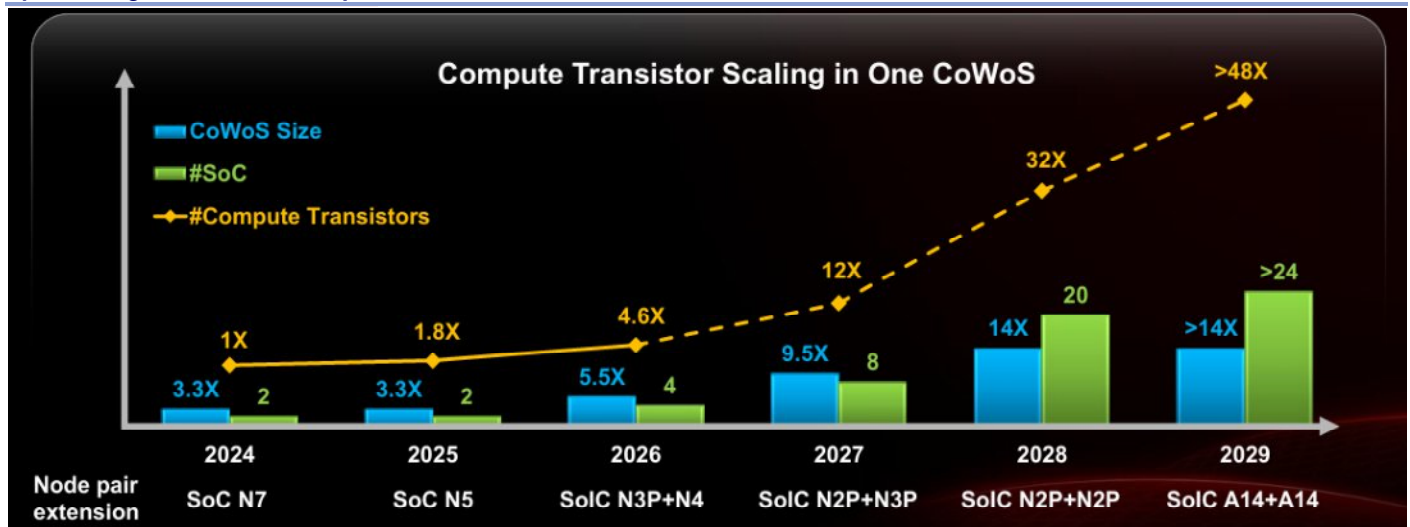
Source: SinoPac, TSMC

TSMC adopts new technology to meet clients' needs

Importantly, future AI compute scaling is not only driven by larger CoWoS size, but also by the combination of advanced logic and SoIC 3D stacking. According to TSMC's roadmap, the number of SoC dies integrated in one CoWoS package can increase from two in 2024-25 to four in 2026, eight in 2027, 20 in 2028 and more than 24 in 2029, while compute transistors in one CoWoS could increase by more than 48x from 2024 to 2029. This indicates that CoWoS scaling is evolving from a relatively simple 2.5D integration of logic die plus HBM towards a more complex system-level integration platform, combining larger interposers, more logic dies and 3D-stacked compute structures.

Figure 14

System integration scales AI compute



Source: SinoPac, TSMC

Emerging packaging technologies boost AI scalability

SoIC stacks chips vertically

However, this also means conventional CoWoS faces greater manufacturing and cost challenges. As package size continues to expand, the silicon interposer becomes increasingly large relative to a 12-inch wafer, creating pressure on wafer utilisation, yield, warpage and cost. Therefore, while CoWoS should remain the mainstream AI accelerator packaging platform, simply scaling package size may become increasingly inefficient. This is driving the emergence of additional solutions such as CoPoS (Chip-on-Panel-on-Substrate)/panel-level packaging, which aim to improve scalability and cost efficiency beyond wafer-based interposers, while SoIC further enhances compute density through vertical die stacking.

SoIC: vertical stacking to improve compute density

SoIC is TSMC's 3D silicon stacking technology under its 3D Fabric platform. Unlike CoWoS, which mainly scales by placing logic dies and HBM side by side on a 2.5D interposer, SoIC enables vertical stacking of chiplets through 3D inter-chip integration. The key enabler is more advanced die-to-die bonding. Compared with conventional micro-bump connections, hybrid bonding removes the need for solder bumps and directly bonds copper pads and dielectric layers between dies, enabling much finer bonding pitch, higher interconnect density, shorter signal path, lower latency and better power efficiency. TSMC's SoIC-X uses a bumpless bonding approach for high-performance applications, while SoIC-P adopts a micro-bump-based scheme as a more cost-effective 3D stacking solution. We expect TSMC's SoIC capacity to reach ~20k by end-2027, with initial adoption likely focused on high-end AI/HPC products.

SoIC integrates both homogeneous and heterogeneous chiplets into a single SoC-like chip

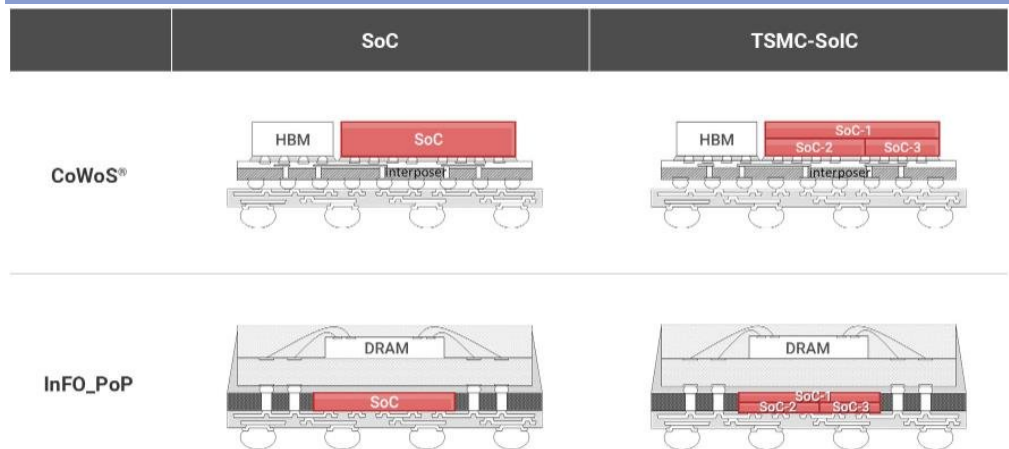
SoIC use hybrid bonding to achieve smaller pitch

Future AI accelerator scaling is moving to 2.5D + 3D integration

SoIC-based CoWoS should still increase back-end process complexity

Figure 15

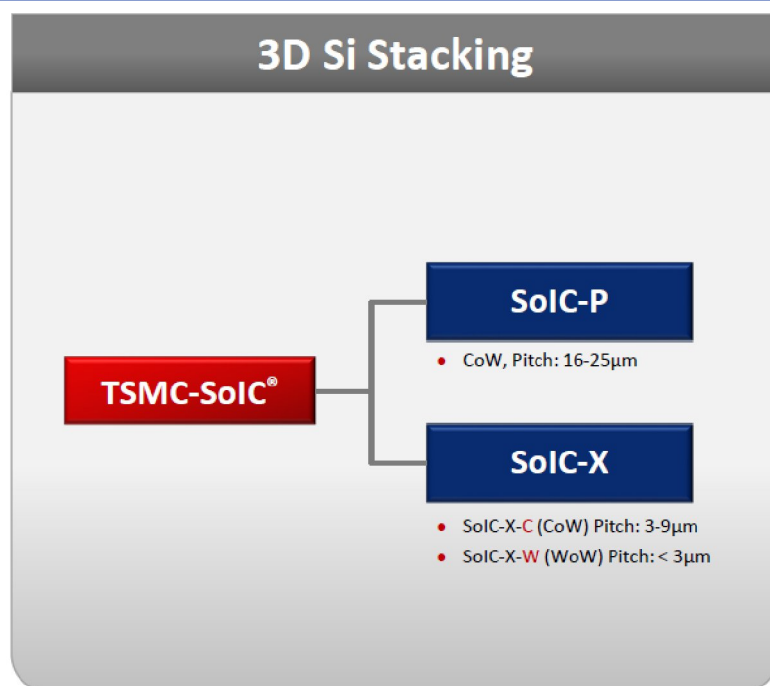
TSMC holistic 3D system integration



Source: SinoPac, TSMC

Figure 16

Comparison between SoIC-P and SoIC-X



Source: SinoPac, TSMC

Leveraging the technology of SoIC, future AI accelerator scaling is moving from pure 2.5D horizontal expansion towards hybrid 2.5D + 3D integration. CoWoS remains critical for integrating logic dies with HBM, while SoIC helps increase compute density by stacking logic dies vertically within the same package footprint. However, SoIC also introduces higher technical and cost challenges, including bonding accuracy, known-good-die control, thermal management, design complexity and yield risk.

From an OSAT perspective, SoIC itself is likely to remain more controlled by TSMC given its tight integration with front-end process and bonding technology.

TSMC developed CoPoS to overcome CoWoS limits

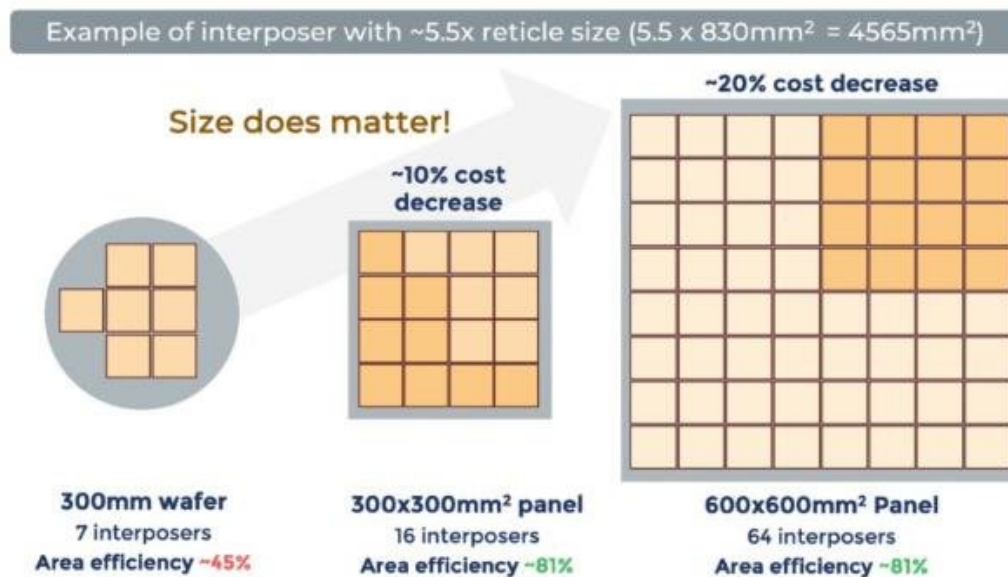
However, broader adoption of SoIC-based CoWoS should still increase back-end process complexity, including advanced assembly, thermal management, final test and system-level validation, creating incremental opportunities for OSAT and testing vendors around the broader advanced packaging ecosystem.

CoPoS/FOPLP: panel-level scaling to address wafer-based limitations

CoPoS/FOPLP is emerging as one of the key next-generation solutions to address the scaling limitations of wafer-based CoWoS. Compared with conventional CoWoS, which integrates logic dies and HBM on a wafer-based interposer, CoPoS moves part of the process to a larger rectangular panel format, aiming to improve area utilisation, unit output and cost efficiency. TSMC is developing CoPoS as a panelised extension of CoWoS for AI/HPC applications, with a mini line expected in 2027 and ready for mass production in 2028. We expect TSMC's CoPoS capacity to reach 3-5/10-15kwpm by end-2027/28.

Figure 17

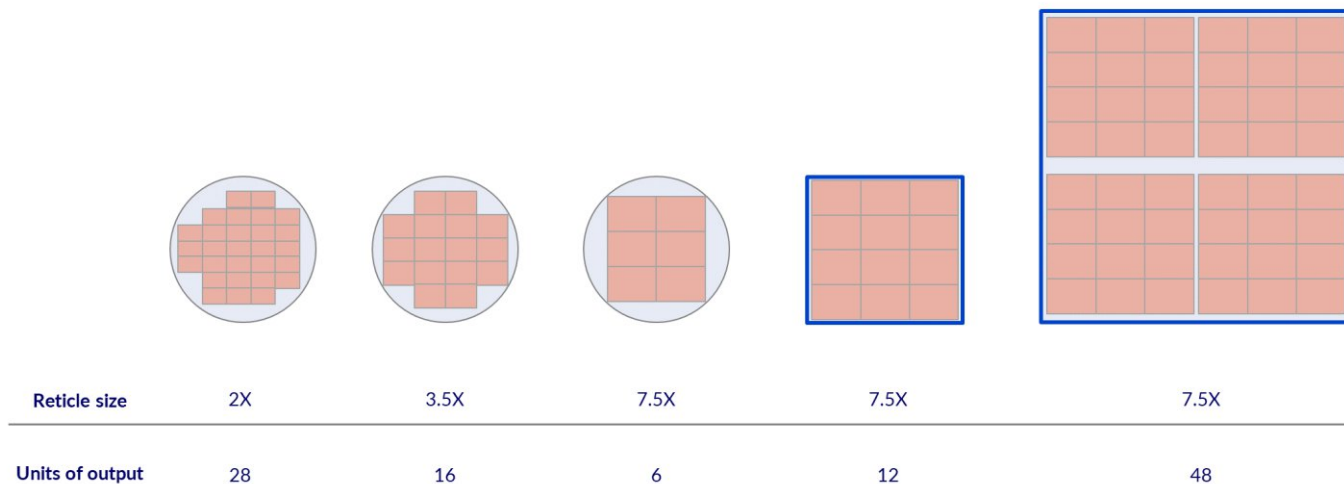
FOPLP provides higher utilisations



Source: SinoPac, Yole

Figure 18

Reticle size and unit comparison



Source: SinoPac, ASE

Panel-level packaging players target future HPC

Beyond TSMC, Powertech and Innolux are also early participants in panel-level packaging. Powertech has been developing FOPLP with a larger panel format of 510x515mm, while Innolux is leveraging its panel manufacturing expertise. However, most projects at the current stage remain non-leading applications such as PMIC, RF, and consumer-related applications such as gaming consoles, while HPC-related projects are more yield- and performance-sensitive, which takes a longer time. ASE is also developing its FOPLP technology with two sizes of 310x310mm and 600x600mm, and the company believes the 310x310mm process will be faster with expected small volume in 2027, for AI/HPC applications.

Figure 19

Main players in Panel-level packaging market

Company	Location	L/S (μm)	Panel size (mm ²)	Application	Mass production
TSMC	Taiwan	3/3	310*310	AI	2028
Samsung	South Korea	7/8	510*415	PMIC, AP	In production
Powertech	Taiwan	8/10, 2/2	515*510	PMIC, CPU/gaming console	Non-leading products in production, High-end in end-2026
ASE	Taiwan	8/8, 2/2	310*310	AI/HPC	Qualification in 2026 and likely small volume in 2027;
		8/8	600*600	Non-leading applications	Evaluating
Amkor	Portugal	> 8/8	650*650	PMIC, RF	In production
Innolux	Taiwan	> 8/8	700*700	PMIC, RF	In production;
		2/2	620*750		Evaluating
Nepes	South Korea	> 8/8	300*300 600*600	PMIC, RF	In production

Source: SinoPac, TrendForce

FOPLP faces cost and specification challenges

Despite increasing efforts from various industry players, FOPLP technology still faces several challenges. As it is still in the early stages of development, one of the major obstacles is the significant investment needed to retool assembly lines and adapt coating technologies for the larger, rectangular form factors. Besides, current FOPLP production processes generally use looser redistribution layer (RDL) pitches and lower layer counts, particularly for consumer and wearable products. More advanced specifications are still in the pilot phase and have not yet been widely adopted.

Testing is another hidden bottleneck

Semiconductor testing is shifting from a minor cost item to a critical capacity constraint driven by the complicated design of AI chips. Chiplet-based designs integrate more dies into a single package, with each die requiring screening before assembly and more comprehensive validation after integration. As a result, testing demand is rising not only from higher AI chip shipments, but also from longer test time, more test insertions, and increasing adoption of burn-in and system-level test. With leading providers already running at high utilisation, we believe testing will become an increasingly important revenue and margin driver for the OSAT sector for leading providers such as KYEC.

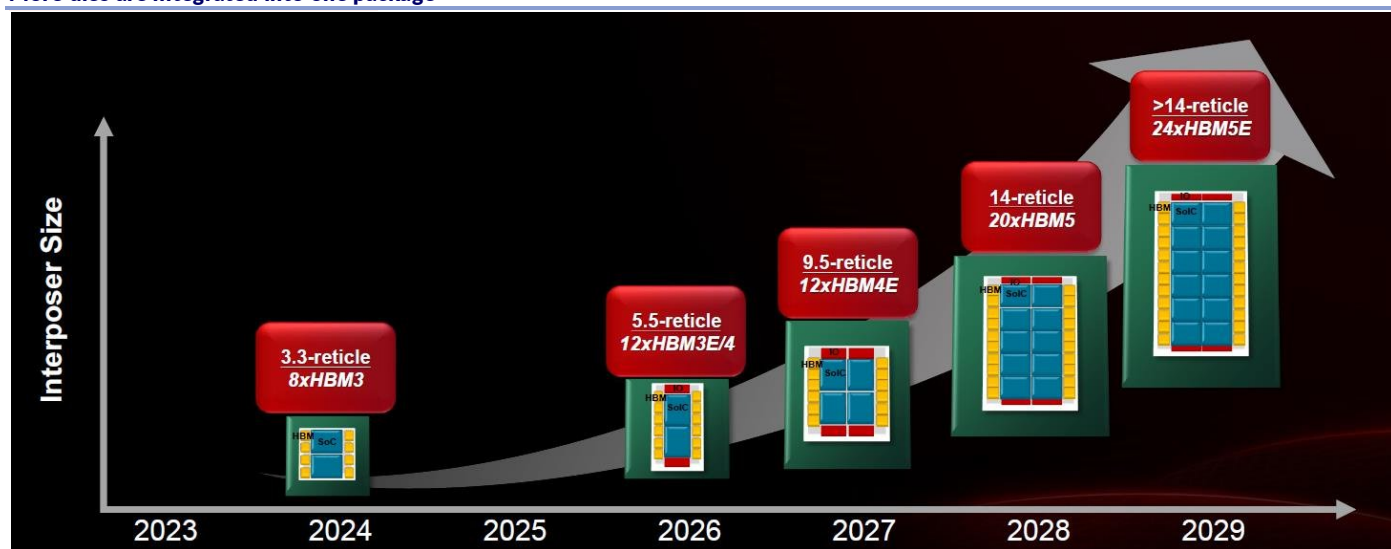
Integration multiplies test content

As AI chips shift towards chiplet-based architectures, a single package now integrates more logic, I/O and HBM dies to scale performance beyond monolithic limits. While the packaging implications were discussed earlier, the same transition also changes the downstream testing burden: more dies create more potential failure points and require more test coverage before and after assembly, setting the stage for longer test time and more complex test flows.

Chiplet design increases the testing requirement

Figure 20

More dies are integrated into one package



Source: SinoPac, TSMC

Testing content is increasing

The sharp increase in the number of dies integrated into a single package is mechanically multiplying testing demand. Densely stacked, high-power AI and HPC devices generate greater heat and operate at higher voltages, lengthening and complicating the testing process. Testing, historically around 2% of total chip cost, is now rising towards a materially larger share of the manufacturing cost mix as chiplet architectures, advanced packaging integration, and system-level test (SLT) add more test insertions and extend test programs.

Chiplet-based architectures require more complex and extended testing flows

Chiplet-based architectures require more complex and extended testing flows than monolithic designs, with each die undergoing wafer-level known-good-die (KGD) screening and often partial-assembly testing before final test to safeguard yield. This introduces multiple additional test insertions, such as burn-in and system-level test, which significantly increase overall testing costs, while defects found later in the process drive scrap losses that rise sharply towards final assembly.

Clearest structural change is the mainstream adoption of SLT

Test flow extends

From a process perspective, the clearest structural change is the mainstream adoption of SLT. Historically, SLT played a complementary, sample-based role, validating a limited subset of devices under end-user operating conditions to confirm system performance. However, as advanced high-density multi-die packages become more complex and operate at higher speeds and power, failures increasingly occur in-system despite passing standalone tests. Manufacturers now run SLT broadly alongside traditional standalone testing, ensuring products meet specifications in real-world environments. The Blackwell test flow reflects this trend: final test, burn-in, a second final test, and then SLT, which is materially longer than the prior Hopper generation. Burn-in requirements are also rising; for example, chamber capacity for Blackwell-class devices has been expanded from 600W to 1kW to accommodate higher power densities.

Figure 21

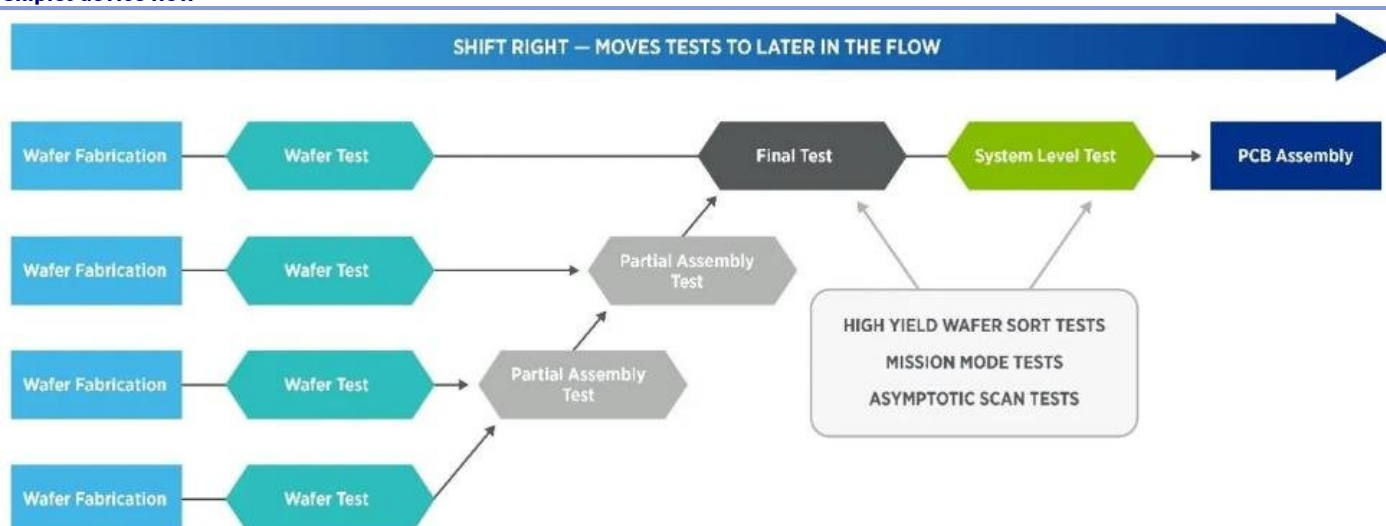
Monolithic device flow



Source: SinoPac, Teradyne

Figure 22

Chiplet device flow



Source: SinoPac, Teradyne

Testing time increasing

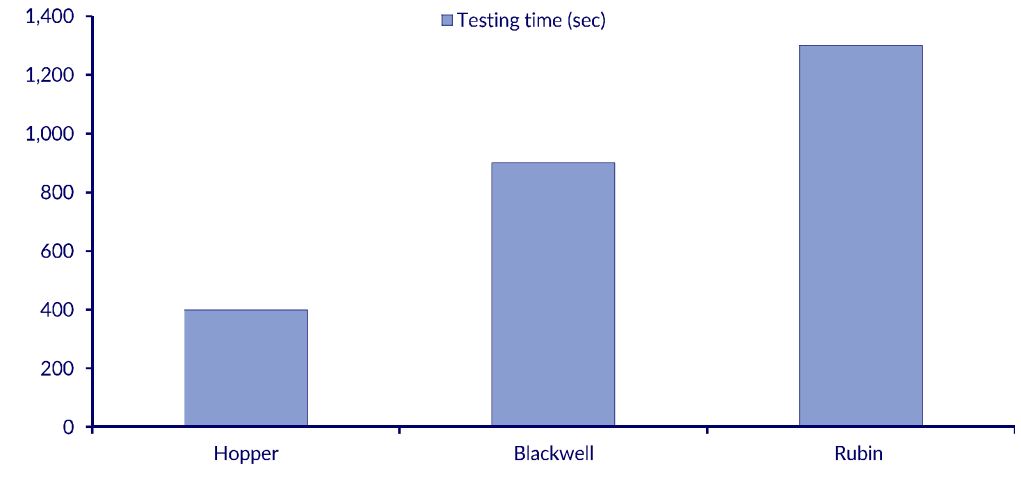
On testing time, Nvidia's FT times have lengthened significantly as programs grow more complex, more than doubling from Hopper to Blackwell and potentially rising another 30-50% with the Rubin platform in 2026, followed by a further step-change with Rubin Ultra in 2027, which will double GPU and HBM counts. Strong AI GPU unit demand and rich FT content are set to drive sustained revenue growth for testing suppliers. While AMD and ASIC vendors such as Google's TPUs currently have shorter test times, the move towards larger packages, increased GPU/HBM integration, and more complex architectures from 2026 onwards is likely to

Longer testing aligns with AI chip roadmap

materially expand their testing capacity needs. For example, MediaTek's TPU Zebrafish, ramping in late 2026, may adopt burn-in and two FT rounds, with other ASIC projects expected to follow.

Figure 23

Testing time increasing with higher design complexity and node migration



Source: SinoPac

Strong LEAP momentum in 2026 beyond

Assume coverage at Outperform

Stock selection

ASE (3711 TT - NT\$539 - O-PF - Target price NT\$161→670)

We expect ASE to remain a key beneficiary of rising advanced packaging demand in 2026 and beyond, with its LEAP platform supported by oS outsourcing and spillover orders from TSMC. With TSMC's CoWoS capacity expanding from 120-125kwpm by end-2026 to 190-200kwpm by end-2027, we expect ASE to see a similar growth trajectory in oS outsourcing and CP demand. For full-process FoCoS, following initial adoption by AMD Venice CPU and AWS Trainium3 this year, Nvidia Vera CPU, Broadcom networking ASICs and AMD consumer-grade products are also evaluating ASE's FoCoS process, providing another growth driver going forward. Moreover, as the industry's technology focus gradually shifts towards CoPoS and CPO, ASE is also developing related solutions and working with key ecosystem partners, positioning the company to capture the next wave of advanced packaging demand beyond current CoWoS outsourcing.

ASE guided that 2026 LEAP revenue could reach at least US\$3.5bn (and US\$5.5bn in 2027), more than doubling from US\$1.6bn in 2025. As ASE's capacity continuously expands, we anticipate its ATM revenue to grow step by step, with revenue YoY growth to reach +31/+28% in 2026/27SP. Supported by surging demand, ASE's utilisation is likely to remain at high levels, with testing UTR in the high-70% range and leading-edge business running at full capacity, setting the stage for margin expansion. Management expects ATM segment GPM to return to the structural margin range (24-30%) by the end of 2026, underpinned by higher utilisation rate and ramp up of margin-accretive leading-edge business. With AI demand scaling, we believe ASE is well-positioned to cement its leadership in the advanced packaging market and secure rising testing opportunities, delivering sustained growth and profitability over the long term. We assume coverage at Outperform with a TP of NT\$670 (previously NT\$161), based on 26x 2027SP EPS.

Figure 24

Quarterly P&L

(NT\$m)	1Q26	2Q26SP	3Q26SP	4Q26SP	1Q27SP	2Q27SP	3Q27SP	4Q27SP	2025	2026SP	2027SP	2028SP
Sales	173,662	188,424	205,085	217,408	211,064	227,581	253,083	263,907	645,388	784,578	955,634	1,101,630
Gross profit	34,850	39,102	43,353	47,147	46,557	50,434	58,519	62,596	114,193	164,452	218,106	263,850
Opex	17,318	18,277	18,458	19,567	18,996	19,800	20,247	21,904	63,437	73,619	80,946	92,227
Operating profit	17,532	20,825	24,895	27,581	27,562	30,635	38,272	40,691	50,756	90,832	137,160	171,624
Non-OP	668	497	497	497	497	497	497	497	545	2,160	1,989	1,989
Pretax profit	18,200	21,322	25,392	28,078	28,059	31,132	38,770	41,189	51,301	92,993	139,149	173,613
Net profit	14,148	16,944	20,493	22,565	22,647	25,023	31,393	33,219	40,658	74,149	112,282	140,160
EPS (NT\$)	3.26	3.90	4.72	5.20	5.22	5.76	7.23	7.65	9.37	17.08	25.86	32.29
Growth												
Sales QoQ	(2%)	9%	9%	6%	(3%)	8%	11%	4%				
Sales YoY	17%	25%	22%	22%	22%	21%	23%	21%	8%	22%	22%	0%
EPS QoQ	(4%)	20%	21%	10%	0%	10%	25%	6%				
EPS YoY	87%	125%	89%	53%	60%	48%	53%	47%	25%	82%	51%	0%
Margin												
Gross margin	20.1%	20.8%	21.1%	21.7%	22.1%	22.2%	23.1%	23.7%	17.7%	21.0%	22.8%	24.0%
Opex	10.0%	9.7%	9.0%	9.0%	9.0%	8.7%	8.0%	8.3%	9.8%	9.4%	8.5%	8.4%
OP margin	10.1%	11.1%	12.1%	12.7%	13.1%	13.5%	15.1%	15.4%	7.9%	11.6%	14.4%	15.6%
Non-OP	0.4%	0.3%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.1%	0.3%	0.2%	0.2%
Net margin	8.1%	9.0%	10.0%	10.4%	10.7%	11.0%	12.4%	12.6%	6.3%	9.5%	11.7%	12.7%

Source: SinoPac

Figure 25

Sales mix assumptions

Sales (NT\$m)	2025	2026SP	2027SP	2028SP
ATM	385,313	503,077	642,142	756,772
EMS	257,194	278,676	310,712	342,122
Others	2,881	2,826	2,780	2,735
Total	645,388	784,578	955,634	1,101,630

YoY	2025	2026SP	2027SP	2028SP
ATM	20%	31%	28%	18%
EMS	(5%)	8%	11%	10%
Others	(4%)	(2%)	(2%)	(2%)
Total	8%	22%	22%	15%

Sales mix	2025	2026SP	2027SP	2028SP
ATM	60%	64%	67%	69%
EMS	40%	36%	33%	31%
Others	0%	0%	0%	0%
Total	100%	100%	100%	100%

Source: SinoPac

Figure 26

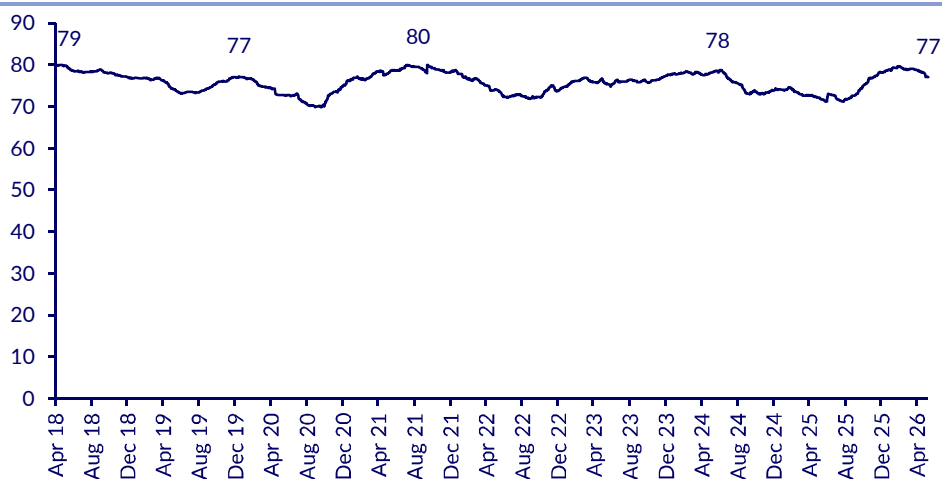
Target price change

	Target PE (x)	Valuation period	EPS (NT\$)	Target price (NT\$)
Old	13	2026SP	12.4	161
New	26	2027SP	25.9	670

Source: SinoPac

Figure 27

QFII holdings

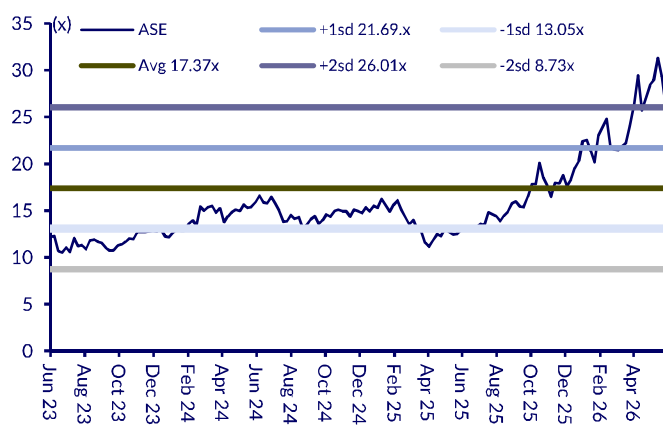


Source: SinoPac

Foreign investors hold 77%

Figure 28

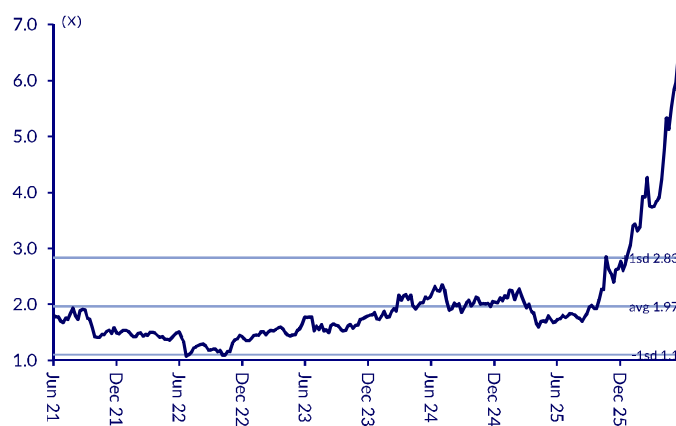
Forward PE



Source: SinoPac

Figure 29

Forward PB



Source: SinoPac

AI chip proxy with high exposure to GPUs and ASICs

Assume coverage at Outperform

KYEC (2449 TT - NT\$272 - O-PF - Target price NT\$142→360)

We view KYEC as one of the best AI chip proxies in Taiwan OSAT, given its broad exposure to most leading AI chip programs, with sole-source or dominant-share positions across Nvidia GPU/CPU, Google TPU/CPU and AI5. The structural trend towards longer testing time and more complicated test flows remains intact for AI chips, driven by node migration, larger die sizes and multi-die designs. Although uncertainty around Rubin Ultra's potential design change could weigh on KYEC's valuation in the near term, we remain positive on the continued increase in testing time and testing insertion. Notably, Nvidia's aggressive Feynman architecture roadmap appears unchanged, with higher die counts per GPU and new technologies such as SolC still likely to be adopted. In our view, this should further reinforce KYEC's role as a key beneficiary of rising AI testing complexity, rather than alter the longer-term growth direction.

KYEC guides 10-11% QoQ revenue growth in 2Q26, with GPM expected to remain flat sequentially. Looking ahead, we believe KYEC is set to enter a robust upcycle through 2026-27, driven by a stronger volume ramp-up of AI GPUs and ASICs in coming quarters and a more diversified client base, supporting full utilisation. KYEC also has confidence in maintaining GPM at high levels, supported by rising contribution from leading-edge testing, with advanced nodes expected to account for c.70% of total revenue in 2026. We expect the leading-edge revenue mix to expand steadily, as more complex chip architectures drive longer testing time and additional testing insertions, supporting further profitability improvement. While the stock has underperformed recently due to concerns over Rubin Ultra design changes and Rubin delays, we believe these concerns should start to fade as Rubin gradually ramps from 3Q26, allowing the market to refocus on KYEC's structural AI testing growth. We assume coverage at Outperform with a TP of NT\$360 (previously NT\$142), based on 23x 2027SP EPS.

Figure 30

Quarterly P&L												
(NT\$m)	1Q26	2Q26SP	3Q26SP	4Q26SP	1Q27SP	2Q27SP	3Q27SP	4Q27SP	2025	2026SP	2027SP	2028SP
Sales	10,192	11,345	13,221	15,571	15,943	16,737	18,233	19,218	34,934	50,329	70,131	91,054
Gross profit	4,051	4,489	5,133	6,270	6,427	6,797	7,411	7,994	12,522	19,942	28,630	38,452
Opex	1,279	1,361	1,124	1,168	1,196	1,255	1,367	1,441	3,520	4,932	5,260	5,810
Operating profit	2,772	3,127	4,009	5,102	5,231	5,542	6,044	6,553	9,003	15,010	23,370	32,643
Non-OP	36	30	30	30	30	30	30	30	1,626	126	119	119
Pretax profit	2,808	3,157	4,039	5,131	5,261	5,572	6,074	6,583	10,629	15,136	23,489	32,762
Net profit	2,286	2,526	3,231	4,105	4,209	4,457	4,859	5,266	11,016	12,148	18,791	26,209
EPS (NT\$)	1.87	2.07	2.64	3.36	3.44	3.65	3.97	4.31	9.01	9.94	15.37	21.43
Growth												
Sales QoQ	2%	11%	17%	18%	2%	5%	9%	5%				
Sales YoY	39%	36%	42%	56%	56%	48%	38%	23%	30%	44%	39%	30%
EPS QoQ	2%	10%	28%	27%	3%	6%	9%	8%				
EPS YoY	(47%)	16%	40%	83%	84%	76%	50%	28%	42%	10%	55%	39%
Margin												
Gross margin	39.7%	39.6%	38.8%	40.3%	40.3%	40.6%	40.6%	41.6%	35.8%	39.6%	40.8%	42.2%
Opex	12.5%	12.0%	8.5%	7.5%	7.5%	7.5%	7.5%	7.5%	10.1%	9.8%	7.5%	6.4%
OP margin	27.2%	27.6%	30.3%	32.8%	32.8%	33.1%	33.1%	34.1%	25.8%	29.8%	33.3%	35.8%
Non-OP	0.4%	0.3%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	4.7%	0.2%	0.2%	0.1%
Net margin	22.4%	22.3%	24.4%	26.4%	26.4%	26.6%	26.6%	27.4%	31.5%	24.1%	26.8%	28.8%

Source: SinoPac

Figure 31

Sales mix assumptions				
Sales (NT\$m)	2025	2026SP	2027SP	2028SP
Wafer probe	11,147	14,417	16,243	17,695
Final Test	19,590	31,451	47,290	64,825
Others	434	151	160	160
Total	34,934	50,329	70,131	91,054
YoY				
YoY	2025	2026SP	2027SP	2028SP
Wafer probe	20%	29%	13%	9%
Final Test	59%	61%	50%	37%
Others	(67%)	(65%)	6%	0%
Total	30%	44%	39%	30%
Sales mix				
Sales mix	2025	2026SP	2027SP	2028SP
Wafer probe	32%	29%	23%	19%
Final Test	56%	62%	67%	71%
Others	1%	0%	0%	0%
Total	100%	100%	100%	100%

Source: SinoPac

Figure 32

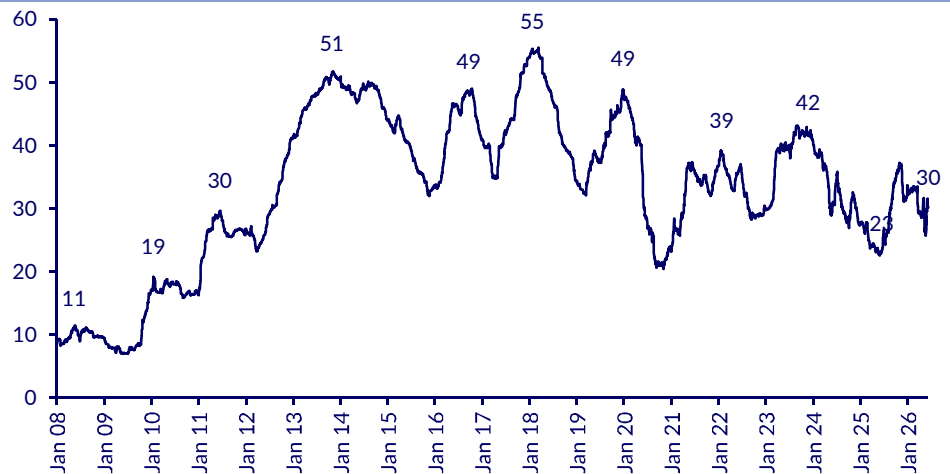
Target price change				
	Target PE (x)	Valuation period	EPS (NT\$)	Target Price (NT\$)
Old	17	2026SP	8.2	142
New	23	2027SP	15.4	360

Source: SinoPac

Foreign investors hold 30%

Figure 33

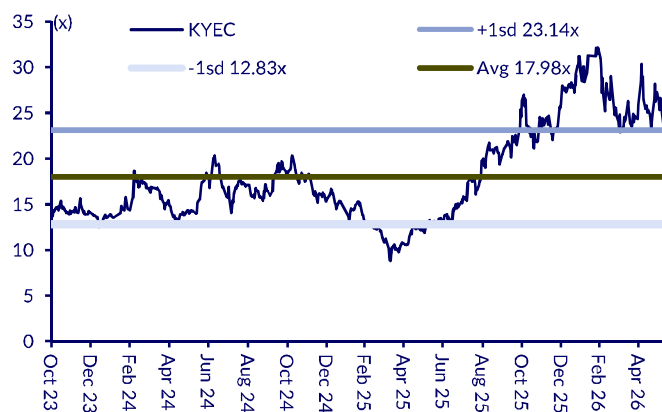
QFII holdings



Source: SinoPac

Figure 34

Forward PE



Source: SinoPac

Figure 35

Forward PB



Source: SinoPac

Memory super cycle and FOPLP beneficiary

Assume coverage at Outperform

Powertech (6239 TT - NT\$303 - HLD→O-PF - TP NT\$125→370)

We expect Powertech to benefit from memory packaging/testing strength, FOPLP commercialisation and longer-term optical packaging opportunities. Memory remains the core earnings base, with DRAM/NAND packaging and testing accounting for around 40-50% of total revenue, supported by strong memory demand driven by AI server demand. For FOPLP, after several years of development, Powertech is finally moving closer to commercialisation, with yield reaching around 95% and constructive customer discussions ongoing. As CoWoS-like packaging demand expands from AI GPUs to server CPUs, gaming consoles and PC CPUs, we see room for Powertech to capture selected consumer-grade products, especially as TSMC capacity remains constrained and Tier-1 OSATs may focus more on higher-end server opportunities. Longer term, its 6k panels/month target provides further upside if utilisation ramps. Powertech is also positioning for optical packaging opportunities, potentially related to Broadcom, which should be viewed as next-generation product optionality rather than near-term revenue contribution.

Powertech expects revenue to grow in mid-single digits QoQ in 2Q26 and continue sequentially in 3Q/4Q, supported by higher UTR, better product mix, cost pass-

through and continued capacity expansion. Management also expects GPM to improve in the coming quarters and reach around the 20% level in 2026, helped by high utilisation, pricing adjustment and production efficiency improvement. The company is currently running at high utilisation, with testing at 85% and packaging at 90%, supported by strong memory and AI demand. For FOPLP, although capacity remains small at 3k panels/month by end-2026, AMD is expected to deliver by year-end, with more meaningful contribution from 2027. Based on management's indication that 6k fully utilised capacity could generate around NT\$3bn monthly revenue, we estimate 3k fully utilised FOPLP capacity could contribute roughly 15-20% of Powertech's current revenue base. We believe Powertech is entering a revenue and margin upcycle, supported by a memory supercycle, AI testing demand and the finally materialised FOPLP opportunity. We assume coverage at Outperform (previously Hold) with a TP of NT\$370 (previously NT\$125), based on 20x 2027SP EPS.

Figure 36

Quarterly P&L

(NT\$m)	1Q26	2Q26SP	3Q26SP	4Q26SP	1Q27SP	2Q27SP	3Q27SP	4Q27SP	2025	2026SP	2027SP	2028SP
Sales	21,314	22,712	24,748	25,659	24,874	27,562	30,424	32,582	74,929	94,433	115,442	126,533
Gross profit	4,144	4,641	5,180	5,428	5,449	6,184	6,944	7,435	12,729	19,393	26,012	29,061
Opex	1,383	1,363	1,485	1,540	1,443	1,599	1,765	1,890	4,597	5,770	6,696	6,959
Operating profit	2,761	3,278	3,695	3,889	4,006	4,586	5,179	5,545	8,132	13,623	19,317	22,101
Non-OP	396	269	269	269	269	269	269	269	660	1,203	1,076	1,076
Pretax profit	3,157	3,547	3,964	4,158	4,275	4,855	5,448	5,814	8,792	14,826	20,392	23,177
Net profit	1,844	2,290	2,656	2,729	2,854	3,227	3,735	3,929	5,536	9,520	13,745	15,433
EPS (NT\$)	2.49	3.09	3.59	3.69	3.85	4.36	5.04	5.31	7.48	12.85	18.56	20.84
Growth												
Sales QoQ	0%	7%	9%	4%	(3%)	11%	10%	7%				
Sales YoY	38%	26%	24%	20%	17%	21%	23%	27%	2%	26%	22%	10%
EPS QoQ	(1%)	24%	16%	3%	5%	13%	16%	5%				
EPS YoY	57%	139%	73%	46%	55%	41%	41%	44%	(18%)	72%	44%	12%
Margin												
Gross margin	19.4%	20.4%	20.9%	21.2%	21.9%	22.4%	22.8%	22.8%	17.0%	20.5%	22.5%	23.0%
Opex	6.5%	6.0%	6.0%	6.0%	5.8%	5.8%	5.8%	5.8%	6.1%	6.1%	5.8%	5.5%
OP margin	13.0%	14.4%	14.9%	15.2%	16.1%	16.6%	17.0%	17.0%	10.9%	14.4%	16.7%	17.5%
Non-OP	1.9%	1.2%	1.1%	1.0%	1.1%	1.0%	0.9%	0.8%	0.9%	1.3%	0.9%	0.9%
Net margin	8.7%	10.1%	10.7%	10.6%	11.5%	11.7%	12.3%	12.1%	7.4%	10.1%	11.9%	12.2%

Source: SinoPac

Figure 37

Sales mix assumptions

Sales (NT\$m)	2025	2026SP	2027SP	2028SP
DRAM	16,876	19,299	22,845	24,756
NAND	18,464	24,243	27,423	29,368
SiP/Module	8,025	9,833	10,482	10,918
Total	74,929	94,433	115,442	126,533

YoY	2025	2026SP	2027SP	2028SP
DRAM	7%	14%	18%	8%
NAND	(12%)	31%	13%	7%
SiP/Module	7%	23%	7%	4%
Total	2%	26%	22%	10%

Sales mix	2025	2026SP	2027SP	2028SP
DRAM	23%	20%	20%	20%
NAND	25%	26%	24%	23%
SiP/Module	11%	10%	9%	9%
Total	100%	100%	100%	100%

Source: SinoPac

Figure 38

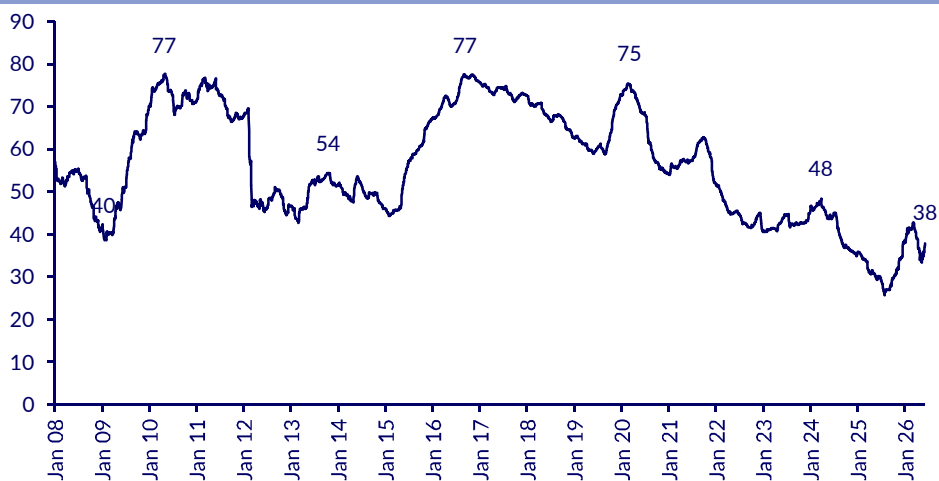
Target price change

	Target PE (x)	Valuation period	EPS (NT\$)	Target Price (NT\$)
Old	11	2026SP	11.9	125
New	20	2027SP	18.6	370

Source: SinoPac

Figure 39

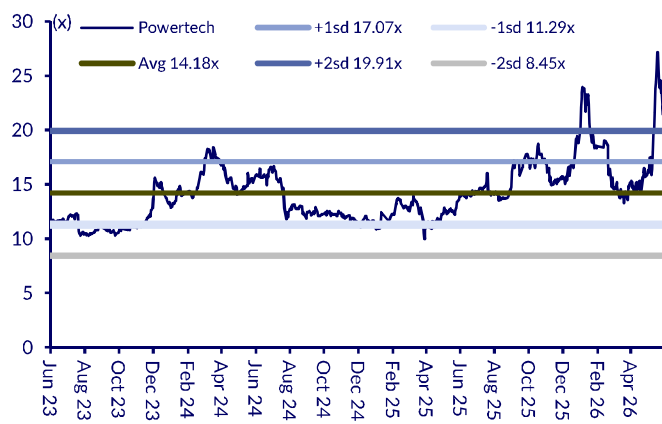
QFII holdings



Source: SinoPac

Foreign investors hold 38%

Figure 40

Forward PE

Source: SinoPac

Figure 41

Forward PB

Source: SinoPac

Investment thesis - ASE Technology Holding Co. Ltd 3711 TT

As TSMC's CoWoS capacity continues to expand, we expect ASE to see a similar growth trajectory in oS outsourcing and CP demand, driving its capacity to maintain at high levels, setting the stage for margin expansion. With AI demand scaling, we believe ASE is well-positioned to cement its leadership in the advanced packaging market and secure rising testing opportunities, delivering sustained growth and profitability over the long term.

Catalysts - ASE Technology Holding Co. Ltd 3711 TT

1) Stronger demand for consumer devices or consumer applications, which could lift wirebond utilisation rates; 2) a smaller-than-expected end-demand correction; and 3) stronger-than-expected AI demand.

Valuation details - ASE Technology Holding Co. Ltd 3711 TT

We use PE as we believe it can better capture the earnings growth of the company. Our target price is based on 26x 2027SP EPS, which is close to +2sd of its five-year average, to reflect rising advanced packaging demand in 2026 and beyond, supported by oS outsourcing and spillover orders from TSMC.

Investment risks - ASE Technology Holding Co. Ltd 3711 TT

1) Weaker demand for consumer devices or consumer applications, which could lead to lower wirebond utilisation rates; 2) a bigger-than-expected end-demand correction; and 3) weaker-than-expected AI demand.

Investment thesis - King Yuan Electronics Co Ltd 2449 TT

We believe KYEC is set to enter a robust upcycle through 2026-27, driven by a stronger volume ramp-up of AI GPUs and ASICs in coming quarters and a more diversified client base, supporting full utilisation. While the stock has underperformed recently due to concerns over Rubin Ultra design changes and Rubin delays, we believe these concerns should start to fade as Rubin gradually ramps from 3Q26, allowing the market to refocus on KYEC's structural AI testing growth.

Catalysts - King Yuan Electronics Co Ltd 2449 TT

1) A faster-than-expected downstream demand recovery; 2) stronger 5G/process node spec migration by clients; 3) stronger-than-expected AI demand.

Valuation details - King Yuan Electronics Co Ltd 2449 TT

We use PE as we believe it can better capture the earnings growth of the company. Our target price is based on 23x 2027SP EPS, which is close to +1sd of its five-year average, to factor in an increased AI revenue contribution and improved GPM, supported by its sole or dominant position in all major AI programs and the structural trend toward longer testing time and more complicated test flows.

Investment risks - King Yuan Electronics Co Ltd 2449 TT

Downside risks include (1) a slower-than-expected downstream demand recovery; (2) weaker 5G/process node spec migration by clients; (3) weaker-than-expected AI demand.

Investment thesis - Powertech Technology Inc 6239 TT

We expect Powertech's revenue growth to be fueled by memory packaging/testing strength, FOPLP commercialisation and longer-term optical packaging opportunities. Management also expects GPM to improve in the coming quarters, helped by high utilisation, pricing adjustments and production efficiency improvement. We believe Powertech is entering a revenue and margin upcycle, supported by the memory supercycle, AI testing demand and the finally materialised FOPLP opportunity.

Catalysts - Powertech Technology Inc 6239 TT

1) Faster-than-expected recovery in HBM, MCU, and PMIC demand; 2) higher contribution from logic business; 3) no further capacity cuts by memory manufacturers.

Valuation details - Powertech Technology Inc 6239 TT

We use a PE-based valuation method as we expect continued profitability growth in 2026/27 thanks to memory packaging/testing strength and FOPLP commercialisation. Our target price is based on 20x 2027SP EPS, which is close to +2sd of its five-year average, to reflect the memory supercycle, AI testing demand and the finally materialised FOPLP business.

Investment risks - Powertech Technology Inc 6239 TT

Downside risks include: 1) a weaker-than-expected recovery in memory demand; 2) lower contribution from logic business; 3) further capacity cuts by memory manufacturers.



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 ARM (N-R)
 ASE (3711 TT - NT\$539.0 - O-PF)²
 Broadcom (AVGO US - US\$392.16 - O-PF)¹
 HiSilicon (N-R)
 Innolux (N-R)
 Intel (N-R)
 King Yuan (2449 TT - NT\$272.0 - O-PF)²
 Marvell (MRVL US - US\$266.88 - O-PF)¹
 MediaTek (2454 TT - NT\$4,155.0 - HC O-PF)²
 Motorola (N-R)
 Nepes Corp (N-R)
 Nvidia (NVDA US - US\$208.19 - HC O-PF)¹
 Powertech (6239 TT - NT\$303.0 - O-PF)²
 Qualcomm (N-R)
 Samsung Electronics (005930 KS - ₩298,000 - O-PF)³
 TSMC (2330 TT - NT\$2,255.0 - HC O-PF)¹

¹ Covered by CLSA Americas; ² Covered by SinoPac; ³ Covered by CLSA

Analyst certification

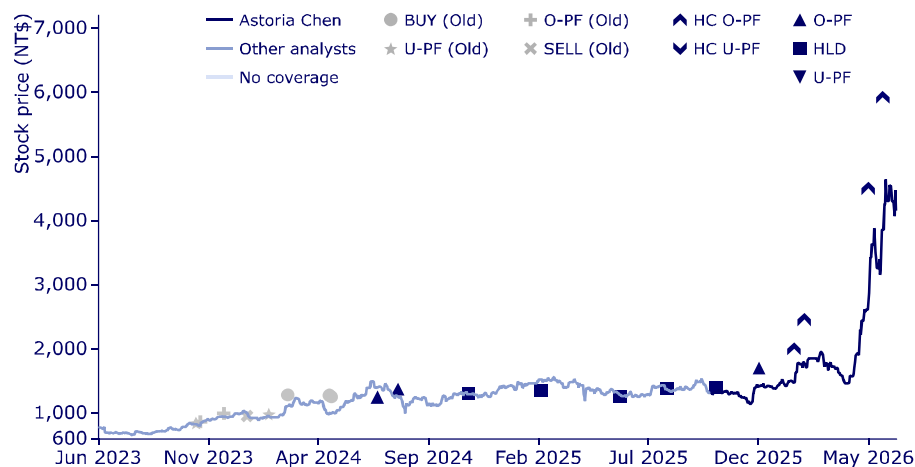
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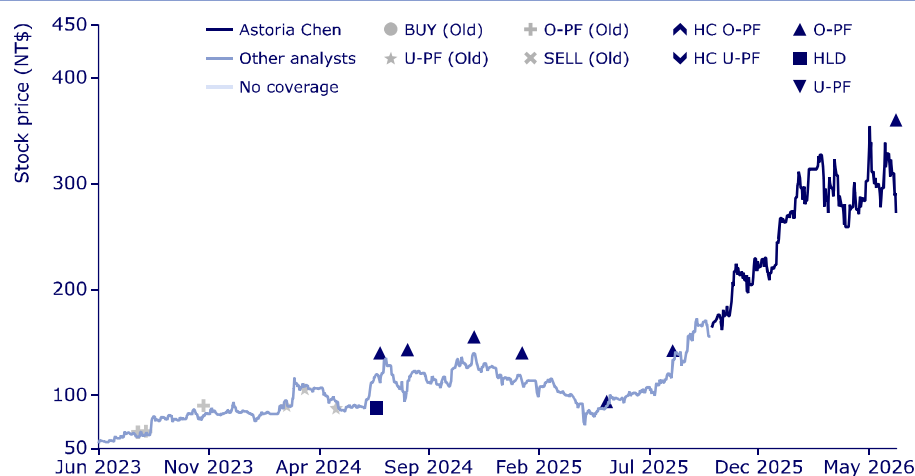
Important disclosures

Recommendation history of MediaTek Inc 2454 TT



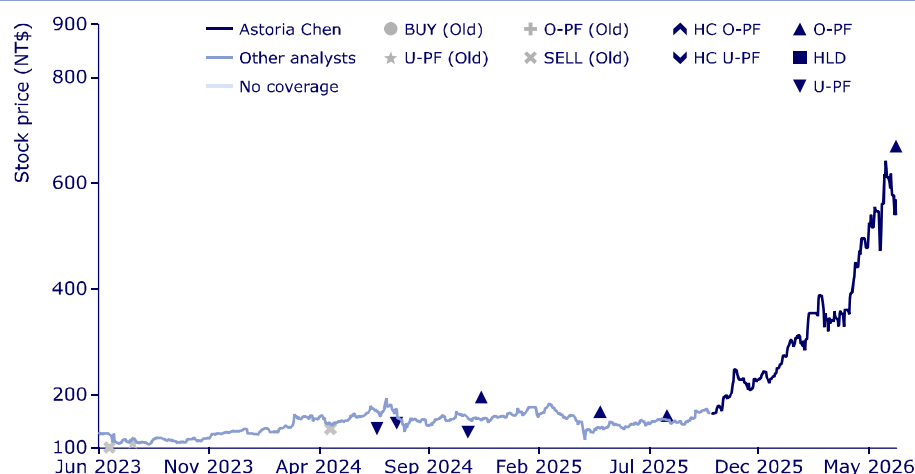
Date	Rec	Target	Date	Rec	Target
23 May 2026	HC O-PF	5,922.00	26 Jul 2024	O-PF	1,370.00
03 May 2026	HC O-PF	4,500.00	28 Jun 2024	O-PF	1,246.00
04 Feb 2026	HC O-PF	2,454.00	26 Apr 2024	BUY	1,246.00
21 Jan 2026	HC O-PF	2,000.00	23 Apr 2024	BUY	1,270.00
04 Dec 2025	O-PF	1,700.00	26 Feb 2024	BUY	1,276.00
06 Oct 2025	HLD	1,400.00	31 Jan 2024	U-PF	980.00
30 Jul 2025	HLD	1,381.00	01 Jan 2024	SELL	952.00
27 May 2025	HLD	1,250.00	30 Nov 2023	O-PF	991.00
08 Feb 2025	HLD	1,350.00	29 Oct 2023	O-PF	860.00
31 Oct 2024	HLD	1,300.00	23 Oct 2023	U-PF	840.00

Note: At 6pm (HKT) on 28 June 2024, we changed our ratings to Outperform (O-PF), Hold (HLD) and Underperform (U-PF); and on 16 December 2024, we added the ratings: High-Conviction Outperform (HC O-PF) and High-Conviction Underperform (HC U-PF). Please see Key to CLSA/SinoPac investment rankings below for details. Source: CLSA, SinoPac

Recommendation history of King Yuan Electronics Co Ltd 2449 TT


Date	Rec	Target	Date	Rec	Target
LATEST	O-PF	360.00	28 Jun 2024	HLD	88.00
08 Aug 2025	O-PF	142.00	03 May 2024	U-PF	88.00
09 May 2025	O-PF	94.00	21 Mar 2024	U-PF	106.00
13 Jan 2025	O-PF	140.00	25 Feb 2024	U-PF	90.00
08 Nov 2024	O-PF	155.00	03 Nov 2023	O-PF	90.00
09 Aug 2024	O-PF	143.00	15 Aug 2023	O-PF	66.00
02 Jul 2024	O-PF	140.00	04 Aug 2023	O-PF	65.00

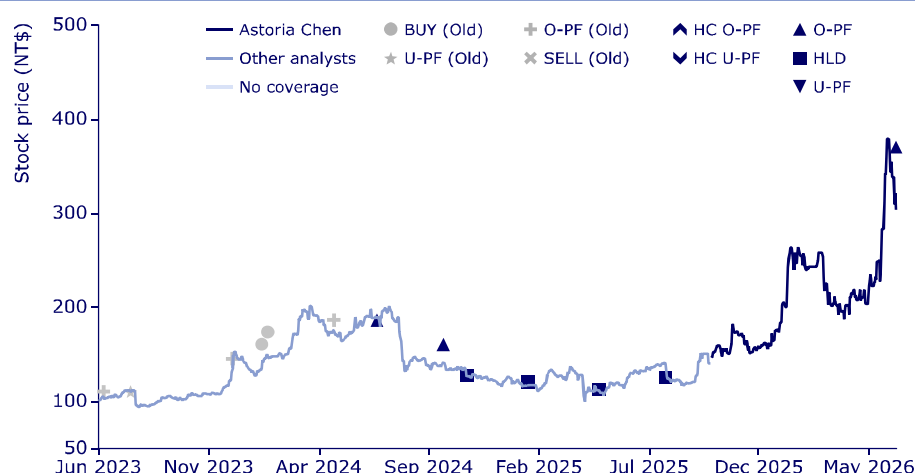
Note: At 6pm (HKT) on 28 June 2024, we changed our ratings to Outperform (O-PF), Hold (HLD) and Underperform (U-PF); and on 16 December 2024, we added the ratings: High-Conviction Outperform (HC O-PF) and High-Conviction Underperform (HC U-PF). Please see Key to CLSA/SinoPac investment rankings below for details. Source: CLSA, SinoPac

Recommendation history of ASE Technology Holding Co. Ltd 3711 TT


Date	Rec	Target	Date	Rec	Target
LATEST	O-PF	670.00	25 Jul 2024	U-PF	146.00
31 Jul 2025	O-PF	161.00	28 Jun 2024	U-PF	136.00
30 Apr 2025	O-PF	168.00	25 Apr 2024	SELL	136.00
18 Nov 2024	O-PF	196.00	28 Jul 2023	U-PF	110.00
31 Oct 2024	U-PF	129.00	26 Jun 2023	SELL	100.00

Note: At 6pm (HKT) on 28 June 2024, we changed our ratings to Outperform (O-PF), Hold (HLD) and Underperform (U-PF); and on 16 December 2024, we added the ratings: High-Conviction Outperform (HC O-PF) and High-Conviction Underperform (HC U-PF). Please see Key to CLSA/SinoPac investment rankings below for details. Source: CLSA, SinoPac

Recommendation history of Powertech Technology Inc 6239 TT



Date	Rec	Target	Date	Rec	Target
LATEST	O-PF	370.00	30 Apr 2024	O-PF	186.00
29 Jul 2025	HLD	125.00	30 Jan 2024	BUY	173.00
29 Apr 2025	HLD	112.00	22 Jan 2024	BUY	160.00
21 Jan 2025	HLD	120.00	13 Dec 2023	O-PF	145.00
29 Oct 2024	HLD	127.00	25 Jul 2023	U-PF	110.00
27 Sep 2024	O-PF	160.00	18 Jun 2023	O-PF	110.00
28 Jun 2024	O-PF	186.00			

Note: At 6pm (HKT) on 28 June 2024, we changed our ratings to Outperform (O-PF), Hold (HLD) and Underperform (U-PF); and on 16 December 2024, we added the ratings: High-Conviction Outperform (HC O-PF) and High-Conviction Underperform (HC U-PF). Please see Key to CLSA/SinoPac investment rankings below for details. Source: CLSA, SinoPac

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Key to CLSA/SinoPac investment rankings: From 16 December

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